Solutions
QA : Numbers - 03

CEX-Q-0221/25

1	1	**2**	1	**3**	3	**4**	4	**5**	1	**6**	2	**7**	3	**8**	3	**9**	3	**10**	1
11	2	**12**	1	**13**	2	**14**	4	**15**	4	**16**	4	**17**	4	**18**	1	**19**	3	**20**	3

1. 1 Required answer = unit digit of (3 × 7 × 3 × 1 × 8 × 6) = 4.

2. 1

3. 3 Unit digit of $(8267)^{153} \times (341)^{72}$
$= \ldots 7^1 \times \ldots 1^2 = 7$.

4. 4 Unit place digit = Unit place digit of (4 × 1 × 9 × 8)
Therefore, unit digit of the number = 8.

5. 1 LCM of cyclicities of 1, 2, 3, … 9 is 4.
∴ If A is a single digit, then last digit of A^{4n+1} will be A only.
Therefore, last digits are 1, 2, 3, 4 … 9 respectively.
Hence, last digit of the sum is 5.

6. 2 $N = 7^{95} - 3^{58}$
The cyclicity of 7 and 3 is 4.
So last digit of $7^{95} - 3^{58}$ will be the same as the last digit of $7^3 - 3^2 = 34^3 - 9 = 334$
Hence, last digit of the number is 4.

7. 3 Any power of 5 when divided by 4 gives a remainder of 1.
Here the power of 3 is itself a power of 5 and will give a remainder of 1 when divided by 4.
The last digit of the number will be $3^1 = 3$.
And hence, last digit of the given number is 3 + 1 = 4.

8. 3 All even powers of 3 are of the form 4n + 1
∴ Last digit of first term is 2 also all odd powers of 3 are of the form 4n + 3
∴ Last digit of the term is 8
{By cyclicity of last digit}
Thus, the last digit of expression is 2 – 8 = 4

9. 3 The last two digits of any number in the form of 7^{4n} will always be equal to 01.
For example: $7^4 = 2401$ and $7^8 = 5764801$.

10. 1 To find the units digit of $(743)^{85} - (525)^{37} + (987)^{96}$
We have to consider only the units digit of each number in the expression.
We know that 3 has a cyclicity of 4, 5 has a cyclicity of 1 and 7 has a cyclicity of 4.
⇒ we find the units digit of the above expression as = 3 – 5 + 1 = 4 – 5
(If we subtract we get – 1, which is not possible, so we borrow from the tens digit)
Hence, we get the units digit as = 14 – 5 = 9.

11. 2

12. 1 1 × 5 × 10 × 15 × 20 × 25 × 30 × 35 × 40 × 45 × 50 × 55 × 60
= 1 × 5 × 5 × **2** × 5 × 3 × 5 × **2** × **2** × 5 × 5 × 5 × 3 × **2** × 5 × 7 × 5 × **2** × **2** × **2** × 5 × 9 × 5 × 5 × **2** × 5 × 11 × 5 × 3 × **2** × **2**
To find the number of zeroes at the end of a product, we need to find the number of 5s and number of 2s in pairs.
Here, in the above product the number of pairs of 2s and 5s = 10.
Hence, the required number of zeroes = 10.

13. 2 Given that 15 × 14 × 13 ×..... 3 × 2 × 1
$= 3^m \times n$
There are 5 multiples of 3 till 15, but 9 is 3^2,
hence, $3^m = 3^6 \Rightarrow m = 6$

14. 4 The number of zeroes in a factorial is determined by the highest power of 5 in it.
The highest power of 5 in 1024! = 253.

15. 4

16. 4 $A = 2 \times 4 \times 6 \times 8 \times 10 \times \ldots\ldots\ldots\ldots 198 \times 200$
$A = 2^{100}(1 \times 2 \times 3 \times 4 \times 5 \ldots\ldots\ldots\ldots 100)$
$A = 2^{100}(100!)$
Highest power of 2 in 100!
$$= \left[\frac{100}{2}\right] + \left[\frac{100}{2^2}\right] + \left[\frac{100}{2^3}\right] + \left[\frac{100}{2^4}\right] + \left[\frac{100}{2^5}\right] + \left[\frac{100}{2^6}\right]$$
= 50 + 25 + 12 + 6 + 3 + 1 = 97.
Similarly, highest power of 5 in 100!
$$= \left[\frac{100}{5}\right] + \left[\frac{100}{5^2}\right] = 20 + 4 = 24.$$

$A = 2^{197} \times 5^{24} \times N$, where N is neither a multiple of 2 nor 5.

$B = 5 \times 10 \times 15 \times 20 \times 25 \times 30 ..... \times 50$

$B = 5^{10}(1 \times 2 \times 3 \times .....10)$

$B = 5^{10} \times 10!$

Highest power of 2 in 10! = 5 + 2 + 1 = 8

Highest power of 5 in 10! = 2.

$B = 5^{12} \times 2^{8} \times M$, where M is not a multiple of 2 or 5.

$\frac{A}{B} = 2^{189} \times 5^{12} \times P$, where P is not a multiple of 2 or 5.

Highest power of 10 in $\frac{A}{B}$ = 12.

Number of zeros at the end of $\frac{A}{B}$ = 12.

17. 4 The number of zeroes will be given by the highest power of 5 in the number.
Now 2! and 4! do not contribute any power of 5
Each 6! contributes 1 power of 5
$\therefore$ $6!^{6!}$ contributes 6! powers of 5
Each 8! contributes 1 power of 5
$\therefore$ $8!^{8}$ contributes 8! powers of 5
Each 10! contributes 2 powers of 5
$\therefore$ $10!^{10!}$ contributes 2 × 10! powers of 5
Thus, total number of zeroes = 6! + 8! + 2 × 10!

18. 1

19. 3 We need to find largest possible value of 'n' such that (n!)! divides 15000! for that n! < 15000
Note: 7! = 5040 and 8! = 40320 > 15000
which implies 15000! is not divisible by 40320!
Therefore, maximum value n can take is 7.

20. 3 Let 'N' be the maximum natural number such that 21^N divides 50!.
i.e., $3^N \times 7^N$ divides 50! completely.
Now, maximum power of 3 that can divide 50! is 22.
Similarly, maximum power of 7 that can divide 50! is 8.
But for 21^N to divide 50! both 3^N and 7^N must divide 50!.
So, min(8, 22) = N
$\Rightarrow$ N = 8.